

Random Matrices

June 12, 2026

Definition 0.1. Consider a random matrix with entries

$$\begin{pmatrix} * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{pmatrix}$$

where these entries are random. Let $X_1, \dots, X_n$ be vectors in $\mathbb{R}^d$, where each $X_i = (X_{i1}, \dots, X_{id})$ consists of random variables.

The variance of the i -th component is given by

$$\frac{1}{n} \sum_{i=1}^n X_{ii}^2 \longrightarrow \text{Variance.}$$

The covariance between two components is given by

$$\frac{1}{n} \sum_{i=1}^n X_{i1} X_{i2} \longrightarrow \text{Covariance.}$$

Let X be the data matrix

$$X = \begin{pmatrix} X_{11} & \cdots & X_{1d} \\ \vdots & \ddots & \vdots \\ X_{n1} & \cdots & X_{nd} \end{pmatrix}.$$

The sample covariance matrix is defined as

$$\frac{XX^\top}{n}.$$

Definition 0.2. The eigenvalues λ of a matrix A are defined by the equation

$$Av = \lambda v \quad \text{for } v \neq 0.$$

The trace of a matrix A is given by

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i,$$

where λ_i are the eigenvalues of A .

Remark 0.1. The set of eigenvalues of a random matrix is referred to as the empirical spectral distribution, denoted by

$$\frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i}.$$

The image contains notes about a random matrix with entries that are probabilistic and symmetric properties of its trace.

Consider a matrix A with entries A_{ij} such that:

$$A = \begin{pmatrix} A_{11} & A_{1j} & \cdots & \\ A_{j1} & A_{jj} & \cdots & \\ \vdots & \vdots & \ddots & \\ A_{m1} & A_{mj} & \cdots & A_{mn} \end{pmatrix}$$

where the entries A_{ij} are independent random variables with probability $\frac{1}{2}$ for 1 or -1 , and $A_{ji} = A_{ij}$.

Definition 0.3. For such a matrix A , the following properties hold for its trace and powers:

$$\text{tr}(A) = \sum_i A_{ii} = \sum_i \lambda_i$$

$$\text{tr}(A^2) = \sum_i A_{ii}^2 = \sum_i \lambda_i^2$$

$$\text{tr}(A^k) = \sum_i A_{ii}^k = \sum_i \lambda_i^k$$